

Synthesis of Anthracene-Based Mechanofluorochromic Molecules and Their Differential Mechanochromic Effects Triggered by Nitrogen Atoms

Hao Yu⁺,^[a] Meng Li⁺,^[a] Xiong Chen,^[a] Ningxu Han,^[a] Zihan Xu,^[b] and Ming Wang^{*[a]}

To investigate the influence of nitrogen atoms on the mechanofluorochromic (MFC) properties, three novel anthracene-based MFC compounds were designed and synthesized, and the effects of nitrogen atom position changes on their photophysical properties and MFC properties were deeply explored. The structure of the three compounds maintains the anthracene and enone portions unchanged, with only subtle adjustments made to the position of the nitrogen atom in the

pyridine moiety. Experimental verification revealed that these three anthracene-based compounds exhibit almost identical photophysical properties in solution but display significantly different fluorescence colors and distinct MFC properties in the solid state. The findings of this study not only offer a new perspective for understanding the MFC mechanism of anthracene-based compounds but also provide valuable insights for the design and development of novel MFC molecules.

Introduction

Stimuli-responsive materials, as a forefront research focus in the field of materials science, possess the capability to undergo physical or chemical property changes in response to various external stimuli such as light, electricity, heat, and force.^[1] The development of these materials is profoundly altering our lifestyles and technological applications. Among the reported stimuli-responsive materials, mechanofluorochromic (MFC) materials have garnered significant attention due to their ability to dynamically and reversibly adjust fluorescence color under external mechanical stimulation. This unique feature endows MFC materials with potential widespread applications in areas such as sensors, data storage, and anti-counterfeiting materials.^[2]

Anthracene is an important π -conjugated compound, and anthracene-based compounds typically exhibit excellent photophysical properties, such as high fluorescence quantum efficiency, long-lived excited states, and outstanding photostability.^[3] In recent years, research has increasingly focused on the MFC properties of anthracene-based compounds, with current studies primarily centered on 9,10-distyrylanthracene derivatives.^[4] Although these derivatives demonstrate remarkable performance in the field of mechanochromism, there is relatively limited research on other types of anthracene-based compounds, which hampers the application

and development of anthracene-based compounds in MFC materials. Additionally, the introduction of heteroatoms significantly impacts the photophysical properties and MFC behavior of anthracene-based compounds, with the nitrogen atom being among the most commonly used heteroatoms.^[5] Despite numerous reports on nitrogen-containing anthracene-based MFC materials, detailed analyses on the fine-tuning of nitrogen atoms within the molecular structure and their effects on MFC performance are scarce.^[6] Therefore, expanding the variety of anthracene-based MFC materials and conducting in-depth studies on the effects of fine-tuning nitrogen atoms on MFC behavior are of great significance.

Building upon this, we focus on the design of novel anthracene-based compounds aimed at further enriching the variety of MFC materials. Additionally, we aim to explore the influence of nitrogen atom positioning on the photophysical properties and MFC behavior of anthracene-based compounds. In this study, we designed and synthesized three anthracene-based compounds, **An-1**, **An-2**, and **An-3**, each modified with nitrogen heterocycles and enones (Figure 1). These three compounds are isomers, with the only difference being the position of the nitrogen atom. By adjusting the position of the nitrogen atom, we aim to investigate in-depth the effects of

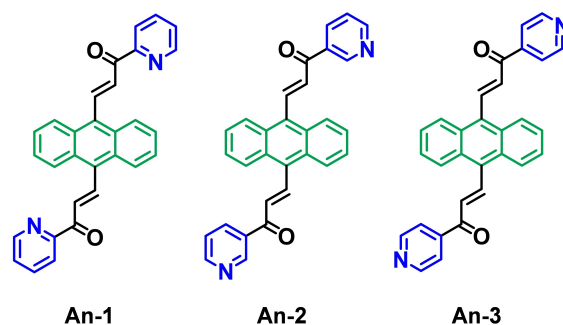


Figure 1. Molecular structures of three anthracene-based compounds.

[a] H. Yu,⁺ M. Li,⁺ X. Chen, N. Han, M. Wang
State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, 130012 Changchun, Jilin, China
E-mail: mingwang358@jlu.edu.cn

[b] Z. Xu
High School Attached to Northeast Normal University, 130117 Changchun, Jilin, China

[⁺] These authors contributed equally to this work.

Supporting information for this article is available on the WWW under <https://doi.org/10.1002/ejoc.202400876>

nitrogen atom variation on the optical properties and MFC behavior of anthracene-based compounds.

Results and Discussion

Compounds **An-1**, **An-2** and **An-3** were obtained by conducting acetylation reactions of anthracene-9,10-dicarbaldehyde with 2-acetylpyridine, 3-acetylpyridine and 4-acetylpyridine, respectively. Detailed synthetic methods and characterizations were provided in the Supporting Information (Scheme S1 and Figures S1–S9). First, we investigated the photophysical properties of these three compounds in the THF solution with a concentration of 10^{-5} M. As shown in Figure 2, the UV-vis absorption spectra of the three compounds exhibited similar absorption features with absorption band centers around 260 nm and 435 nm. The fluorescence spectra of three compounds were recorded at 435 nm excitation. Their emission spectra are also very similar, with emission bands ranging from 475 nm to 650 nm, and the main emission peak centers at around 555 nm. At the excitation wavelength of 435 nm, the absolute emission quantum yields of these three compounds are 1%, 2%, and 2%, respectively, indicating weak fluorescence emission. The similar solution-state photophysical properties of three compounds prove that the position of nitrogen atoms has

minimal impact on their properties in the dispersed state of the solution.

Subsequently, we further investigated the photophysical properties of the three compounds in solution-aggregated states. Figure 3 presents the fluorescence spectra of the three compounds, recorded in solvent mixtures of different proportions of tetrahydrofuran (THF) and water (H_2O). With increasing the water content from THF: H_2O ratios of 10:0 to 1:9, similar trends were observed for all three compounds. Specifically, as the THF: H_2O ratio increased gradually from 10:0 to 3:7, their emission intensities decreased. However, when the ratio reached 1:9, the fluorescence intensity significantly increased. For the phenomenon of initially decreasing and then increasing emission intensity, we speculate that this is due to the gradual increase in solvent polarity as water is added, leading to molecular conformational changes and the formation of weakly emitting twisted intramolecular charge transfer (TICT) excited states. As the proportion of water continues to increase, the intermolecular aggregation increases, restricting the formation of TICT excited states, resulting in an increase in fluorescence intensity.^[7] Furthermore, we recorded the fluorescence intensity of three compounds at different concentrations in both THF and the mixed solvent of THF: H_2O =1:9. Combined with excitation spectra, it was observed that the fluorescence intensity of all three compounds decreased with a reduction in concentration (Figures S12 and S14), both in the dispersed state within the THF solution (Figure S13) and in the aggregated state within the THF: H_2O = 1:9 mixed solvent (Figure S15).

Furthermore, we conducted a solvent-induced chromism study on three compounds, employing four solvents of different polarities: hexane, dichloromethane, tetrahydrofuran, and acetonitrile (Figure S16). The results indicate that the absorption spectra of the three compounds in solvents of varying polarities remain almost unchanged, suggesting that their ground states are not affected. However, the emission spectra of these three compounds in different solvents exhibit noticeable variations, with similar change trends. As the solvent polarity increases, the emission peaks of the three compounds undergo a redshift. The main emission peak centers around 538 nm in the non-polar solvent hexane, while it shifts to around 568 nm in the polar solvent acetonitrile, showing a redshift of approximately 30 nm. This suggests the presence of intramolecular charge transfer phenomena in the three compounds.

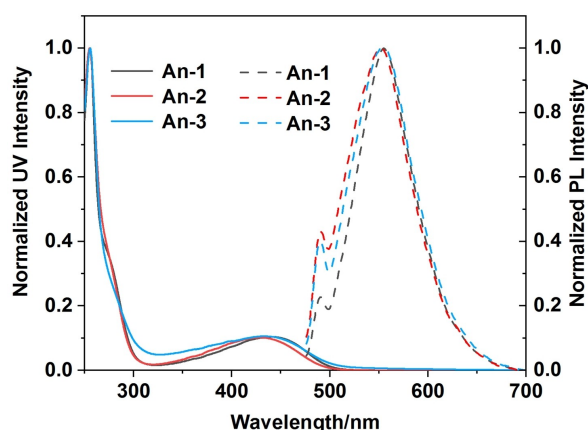


Figure 2. The UV-vis absorption and fluorescence emission spectra of compounds **An-1**, **An-2** and **An-3** ($c = 10^{-5}$ M).

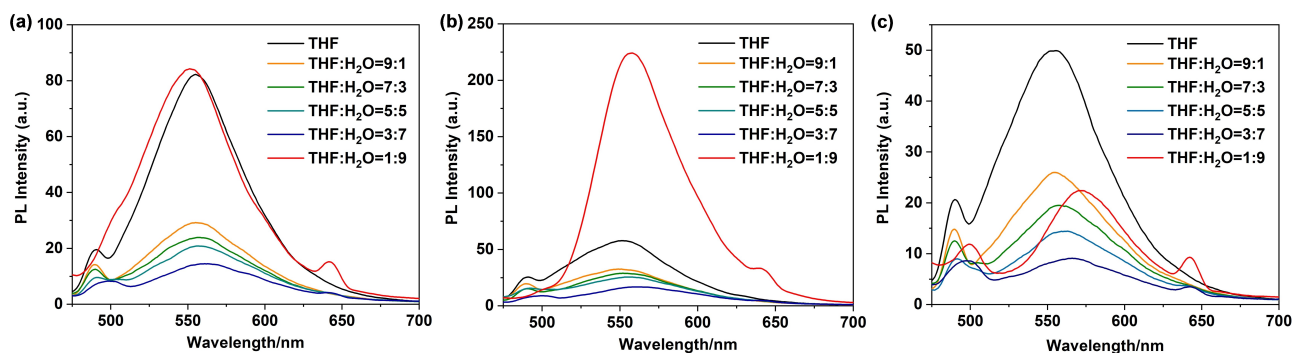


Figure 3. The fluorescence emission spectra of compounds (a) **An-1**, (b) **An-2**, and (c) **An-3** in solutions containing poor solvents are shown ($c = 10^{-5}$ M).

To further investigate the intramolecular charge transfer phenomena of compounds **An-1**, **An-2**, and **An-3**, we conducted density functional theory (DFT) calculations. Using Gaussian 09 software, we optimized the geometries of these three compounds at the B3LYP/6-31G(d,p) level. Through analysis of the frontier molecular orbitals, we further confirmed the intramolecular charge transfer characteristics of **An-1**, **An-2**, and **An-3** (Figure S26). The computational results reveal that the lowest unoccupied molecular orbital (LUMO) is distributed over the entire molecule of the three compounds. Meanwhile, the highest occupied molecular orbital (HOMO) is mainly localized on the anthracene moiety, indicating that the anthracene moiety serves as the primary electron-donating region. This distribution pattern of LUMO and HOMO clearly elucidates the process of charge transfer that can occur in these three compounds. These research findings contribute to a deeper understanding of the electronic structure and photophysical properties of these compounds.

Through the above studies, we have delved into the photophysical behaviors of three compounds in solution and revealed their fluorescence characteristics in poor solvents and different solvent environments. However, experimental results indicate that the photophysical properties of the three compounds in solution are similar, suggesting that the different positions of nitrogen have little influence on their photophysical properties in solution. Next, we characterized the solid-state photophysical properties and MFC properties of all compounds. We thoroughly ground each compound and characterized the solid powders before and after grinding. Figure S18 illustrates the solid-state UV-vis absorption spectra of these three compounds before and after grinding. Comparison reveals that the grinding process did not cause significant changes in the UV-vis absorption of the samples. All three compounds exhibited broad UV-vis absorption bands from 200 nm to 600 nm before and after grinding. Similarly, the compounds also display broad excitation spectra before and after grinding, with consistent peak positions (Figure S19).

However, the intensity of the excitation spectra changes to some extent after grinding, which we speculate is due to changes in intermolecular interactions and molecular packing caused by the grinding process.

Subsequently, we delved into the solid-state photophysical properties and MFC properties of compounds **An-1**, **An-2**, and **An-3** before and after grinding. Fluorescence spectra and fluorescence quantum yields of all samples were tested using an excitation wavelength of 435 nm, with the experimental results shown in Figure 4. The investigation revealed distinct MFC behaviors of these compounds after thorough grinding. For compound **An-1**, featuring a modified 2-pyridyl acceptor moiety, its emission peak underwent a minor redshift of 10 nm (Figure 4b). Both the pre- and post-grinding solid powders exhibited red fluorescence, with fluorescence quantum yields of 6% and 7%, respectively, and the grinding process did not significantly affect the fluorescence intensity of **An-1** (Figure 4a). In contrast, for compound **An-3**, which has a modified 4-pyridyl acceptor moiety, its emission peak experienced a redshift of 14 nm (Figure 4f). The initial solid powder exhibited orange-red fluorescence with a solid-state quantum yield of 34%, while the post-grinding solid powder exhibited red fluorescence with a solid-state quantum yield of 9%. This indicates that grinding not only caused a visible change in the fluorescence color of **An-3** but also significantly affected its fluorescence intensity (Figure 4e). It is worth noting that when the acceptor group was modified to 3-pyridyl in compound **An-2**, the powder exhibited the most significant MFC behavior after grinding. Its emission peak underwent a remarkable redshift of up to 30 nm (Figure 4d). The initial solid powder displayed yellow fluorescence with a solid-state quantum yield of 17%, while the post-grinding solid powder exhibited orange-red fluorescence with a solid-state quantum yield of 9%. This indicates that the grinding process led to a significant change in the fluorescence color of **An-2** and also had a noticeable impact on the fluorescence intensity (Figure 4c). We further investigated the reversibility of the three compounds and found that after

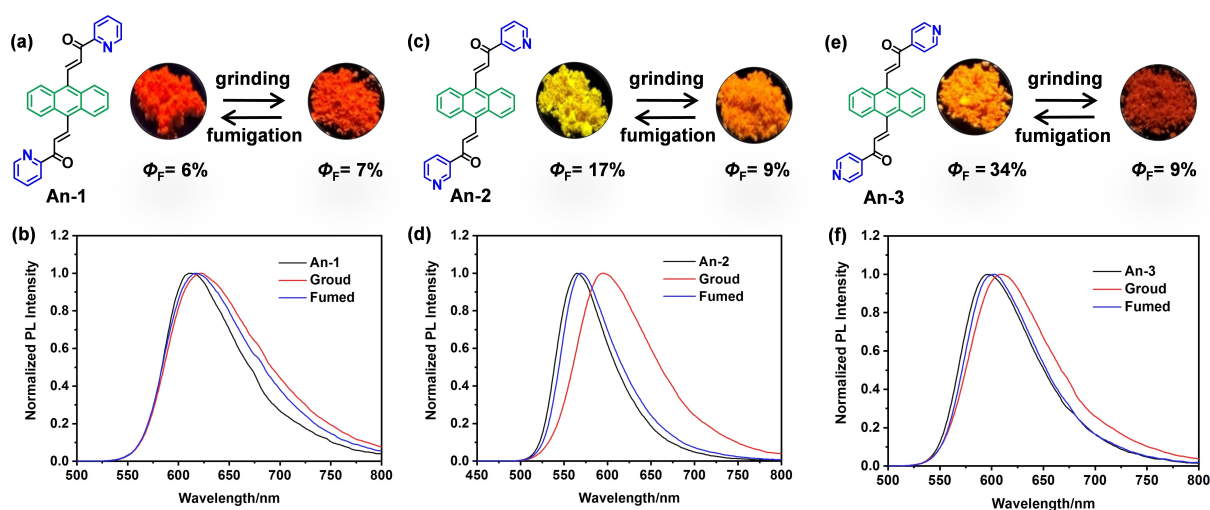


Figure 4. Fluorescence pictures and absolute quantum efficiency of compounds (a) **An-1**, (c) **An-2** and (e) **An-3** before and after grinding; Solid-state fluorescence emission spectra of (b) **An-1**, (d) **An-2** and (f) **An-3** at initial, grinding and solvent fumigation.

fumigation with dichloromethane solvent, the solid-state emission spectra could be effectively restored, indicating their reversible MFC properties. Overall, the three compounds exhibited markedly different solid-state fluorescence emission and MFC properties, highlighting the significant influence of nitrogen at different positions on their solid-state photophysical properties, which contrasts sharply with their similar solution-state photophysical properties. Additionally, we investigated the acid-stimulus response behavior of these three compounds by exposing them to HCl vapor. All three compounds exhibited a red shift in their fluorescence emission. Moreover, when these three compounds were exposed to $\text{NH}_3\cdot\text{H}_2\text{O}$ vapor, the fluorescence color could be reversibly restored (Figure S20).

To gain deeper insights into the mechanistic aspects of the MFC properties of **An-1**, **An-2**, and **An-3**, we conducted powder X-ray diffraction (PXRD) characterization on these three compounds. The initial solids of three compounds exhibited sharp and intense reflections, while the powders after grinding showed varying degrees of peak broadening (Figure S21). This observation indicates that the initial samples possessed an ordered microcrystalline structure, and the samples after grinding still retained some degree of the ordered microcrystalline structure. The alteration in this microcrystalline structure may be closely associated with the formation of MFC properties. Subsequently, we analyzed the thermal behavior of **An-1**, **An-2**, and **An-3** using thermogravimetric analysis (TGA). The TGA analysis revealed that all three compounds undergo thermal decomposition around 320°C , as depicted in Figure S22. This indicates their stability at elevated temperatures, which is crucial for potential applications requiring thermal stability.

To delve deeper into the intricate interplay between the MFC properties exhibited by the three compounds and their respective molecular structures, we conducted in-depth investigations of their molecular aggregation states through single-crystal structure characterization. By diffusing isopropyl ether vapor into chloroform solutions containing each of three compounds, we successfully obtained crystals of each compound. The crystal structures of these compounds and their molecular packing modes are depicted in Figure 5, with crystallographic details provided in Tables S1–S3. Both the anthracene and enone moieties of **An-1**, **An-2**, and **An-3** exhibit significant torsion angles of 48.05° , 45.62° , and 53.04° , respectively, indicating highly distorted non-planar molecular structures. Within the crystal packing of these three compounds, numerous weak interactions were identified (Table 1). **An-1** exhibits 4 $\text{C}\cdots\pi$ interactions, 4 $\text{C}\cdots\text{H}\cdots\text{O}$ interactions, and 4 $\text{H}\cdots\text{H}$ interactions. **An-2** features 16 $\text{C}\cdots\pi$ interactions, 4 $\text{C}\cdots\text{H}\cdots\text{O}$ interactions, and 4 $\text{H}\cdots\text{H}$ interactions. Meanwhile, **An-3** demonstrates 4 $\pi\cdots\pi$ interactions, 4 $\text{C}\cdots\text{H}\cdots\text{O}$ interactions, and 8 $\text{C}\cdots\text{O}$ interactions. We speculate that the highly distorted molecular structures and various weak interactions contribute to the formation of loosely packed molecular assemblies. Under the influence of external mechanical force, the molecular conformations of the three compounds tend towards more planar configurations, inducing intermolecular sliding. As the structure configuration changes, the arrangement of molecular packing becomes more compact, ultimately resulting in significant red

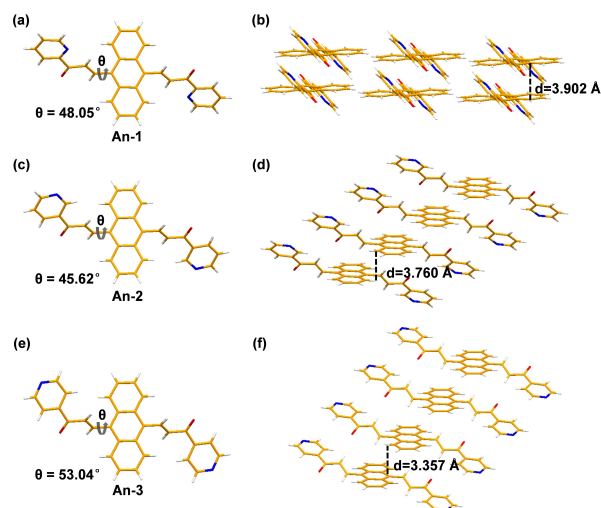


Figure 5. Single crystal structures of compounds (a) **An-1**, (c) **An-2** and (e) **An-3**; Crystal packing diagrams of (b) **An-1**, (d) **An-2** and (f) **An-3**.^[8]

Table 1. The list of various intermolecular interactions for **An-1**, **An-2** and **An-3**.

	An-1 (12 interactions)	An-2 (24 interactions)	An-3 (16 interactions)
$\pi\cdots\pi$			4*3.357 Å
$\text{CH}\cdots\pi$	4*2.684 Å	4*2.776 Å 4*2.783 Å 4*2.849 Å 4*2.875 Å	
$\text{CH}\cdots\text{O}$	4*2.642 Å	4*2.489 Å	2*2.505 Å 2*2.512 Å
$\text{C}\cdots\text{O}$			2*3.196 Å 2*3.129 Å 2*3.180 Å 2*3.080 Å
$\text{H}\cdots\text{H}$	4*2.324 Å	4*2.322 Å	

shifts in the fluorescence emission peaks of the three compounds, accompanied by changes in fluorescence color. By analyzing the crystal packing modes of the three compounds, we can gain deeper insights into the differences in their MFC properties. In the case of **An-1**, the intermolecular anthracene units adopt a face-to-face stacking dimer pattern, resulting in relatively tight molecular packing. Under external force, there is limited space for molecular sliding, hence the small shift in emission peak (Figure 5b). In contrast, the intermolecular packing of **An-2** and **An-3** is looser, allowing for greater molecular sliding space under external force and resulting in larger emission peak shifts (Figure 5d,f). Particularly in the case of **An-2**, the anthracene units are more distantly spaced, resulting in even looser intermolecular packing, which accounts for the maximal fluorescence change observed after grinding. Overall, the position of the nitrogen atom in these three compounds influences the solid-state packing arrangement of the anthracene units, thereby giving rise to distinct MFC properties.

Conclusions

In conclusion, we designed and synthesized three anthracene-based compounds modified with nitrogen heterocycles and enones. These three compounds are isomers, differing only in the position of the nitrogen atom. In the solution state, the photophysical properties of three compounds are remarkably similar, indicating that the change in the position of the nitrogen atom has little influence on their solution-state photophysical properties. However, in the solid state, significant differences are observed in the photophysical properties and MFC properties of the three compounds. Particularly, compound **An-2** exhibits the most pronounced MFC effect, with a red shift in emission peak of up to 30 nm. Single-crystal structure revealed that the different intermolecular forces and molecular stacking patterns caused by the variation in the position of the nitrogen atom are the main reason for their distinct MFC properties. This study demonstrates that the variation in the position of the nitrogen atom significantly influences the solid-state photophysical properties and MFC properties of anthracene-based compounds, providing valuable insights for the molecular design of future MFC materials.

Acknowledgements

We gratefully acknowledge the support from the fellowship of China Postdoctoral Science Foundation (2023M741338 for H.Y.), and Postdoctoral Fellowship Program of CPSF (GZC20230940 for H.Y.).

Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: Mechanofluorochromic materials • Anthracene • Fluorescence • Single crystal

- [1] a) T. Zhang, X.-Y. Lou, X. Li, X. Tu, J. Han, B. Zhao, Y.-W. Yang, *Adv. Mater.* **2023**, *35*, 2210551; b) A. S. Burova, O. V. Venidiktova, M. A. Savelyev, V. A. Barachevsky, T. V. Bukreeva, T. N. Borodina, *Mater. Lett.* **2021**, *303*, 130558; c) L. Wang, T. Yu, Z. Xie, E. Ubba, T. Zhan, Z. Yang, Y. Zhang, Z.

- Chi, *RSC Adv.* **2018**, *8*, 18613–18618; d) J. Du, L. Sheng, Y. Xu, Q. Chen, C. Gu, M. Li, S. X.-A. Zhang, *Adv. Mater.* **2021**, *33*, 2008055; e) Z. Wang, X. Huang, *Chem. Eur. J.* **2022**, *28*, e202200609; f) Z. Wang, X. Wang, S. Cong, J. Chen, H. Sun, Z. Chen, G. Song, F. Geng, Q. Chen, Z. Zhao, *Nat. Commun.* **2020**, *11*, 302; g) M. Y. Livshits, J. J. Rack, *J. Phys. Chem. C* **2016**, *120*, 26459–26464; h) M. Mrinalini, S. Prasanthkumar, *ChemPlusChem* **2019**, *84*, 1103–1121; i) S. Bhui, P. K. Samanta, M. Chakravarty, *Dyes Pigm.* **2022**, *197*, 109944.
- [2] a) M. Xu, B. Zhang, J. Tian, Y. Yang, S. Gao, F. Li, L. Shao, *ChemistrySelect* **2024**, *9*, e202303763; b) B. R. Crenshaw, M. Burnworth, D. Khariwala, A. Hiltner, P. T. Mather, R. Simha, C. Weder, *Macromolecules* **2007**, *40*, 2400–2408; c) Y. Xiao, K. Zheng, N. Zhang, Y. Wang, J. Yan, D. Wang, X. Liu, *Chem. Eur. J.* **2023**, *29*, e202203772; d) S. Bhui, P. Chakraborty, P. Sivasakthi, P. K. Samanta, P. Yogeewari, M. Chakravarty, *J. Mater. Chem. C* **2023**, *11*, 11270–11282; e) M. A. Potopnyk, D. Volyniuk, R. Luboradzki, A. Lazauskas, J. V. Grazulevicius, *Eur. J. Org. Chem.* **2021**, *19*, 2772–2781; f) S. Xue, M. Li, Y. Han, Y. Chen, *Dyes Pigm.* **2021**, *193*, 109484; g) B. Zou, J.-A. Li, Z. Zhai, T. Yu, H. Liu, J. Huang, H. He, B. Xu, H.-W. Jiang, *Eur. J. Org. Chem.* **2024**, *16*, e202400135. h) P. Gayathri, S. Karthikeyan, M. Pannipara, A. G. Al-Sehemi, D. Moon, S. P. Anthony, *Cryst. Growth Des.* **2022**, *22*, 5432–5440; i) S. Takahashi, S. Nagai, M. Asami, S. Ito, *Mater. Adv.* **2020**, *1*, 708–719; j) N. K. Ramya, C. Femina, S. Suresh, D. S. Mohanakumari, R. Krishnan, R. Thomas, *New J. Chem.* **2022**, *46*, 1339–1346; k) Y.-R. Jia, K. Xu, M. Xia, *Dyes Pigm.* **2023**, *214*, 111234.
- [3] a) M. Yoshizawa, J. K. Klosterman, *Chem. Soc. Rev.* **2014**, *43*, 1885–1898; b) X.-H. Zheng, T.-T. Huang, G.-X. Yang, A.-Q. Lin, K. Chen, X. Chen, J.-Y. Li, Q.-X. Tong, *Chem. Eur. J.* **2021**, *27*, 16181–16188; c) P. Devibala, B. Balambiga, P. M. Imran, N. S. P. Bhuvanesh, S. Nagarajan, *Eur. J. Org. Chem.* **2022**, *44*, e202200825; d) H. Yu, J. Shi, M. Li, G. Pan, H. Tong, B. Xu, M. Wang, W. Tian, *Inorg. Chem.* **2022**, *61*, 7231–7237; e) N. Sharma, J. Jung, Y.-M. Lee, M. S. Seo, W. Nam, S. Fukuzumi, *Chem. Eur. J.* **2017**, *23*, 7125–7131; f) K. Santo, K. Uchida, K. Fujimoto, T. Inuzuka, K. Hirakawa, T. Sengoku, M. Takahashi, *Eur. J. Org. Chem.* **2023**, *26*, e202201479.
- [4] a) Y. Dong, B. Xu, J. Zhang, X. Tan, L. Wang, J. Chen, H. Lv, S. Wen, B. Li, L. Ye, B. Zou, W. Tian, *Angew. Chem. Int. Ed.* **2012**, *51*, 10782–10785; b) Y. Dong, J. Zhang, X. Tan, L. Wang, J. Chen, B. Li, L. Ye, B. Xu, B. Zou, W. Tian, *J. Mater. Chem. C* **2013**, *1*, 7554–7559; c) J. Zhao, Z. Chi, Z. Yang, Z. Mao, Y. Zhang, E. Ubba, Z. Chi, *Mater. Chem. Front.* **2018**, *2*, 1595–1608; d) S. Xue, X. Qiu, Q. Sun, W. Yang, *J. Mater. Chem. C* **2016**, *4*, 1568–1578; e) C. Niu, Y. You, L. Zhao, D. He, N. Na, J. Ouyang, *Chem. Eur. J.* **2015**, *21*, 13983–13990.
- [5] a) R. Li, S. Xiao, Y. Li, Q. Lin, R. Zhang, J. Zhao, C. Yang, K. Zou, D. Li, T. Yi, *Chem. Sci.* **2014**, *5*, 3922–3928; b) K. C. Naeem, A. Subhakumari, S. Varughese, V. C. Nair, *J. Mater. Chem. C* **2015**, *3*, 10225–10231; c) X. Zhang, Y.-X. Wang, J. Zhao, P. Duan, Y. Chen, L. Chen, *Chem. Asian J.* **2017**, *12*, 830–834; d) P. Xue, B. Yao, X. Liu, J. Sun, P. Gong, Z. Zhang, C. Qian, Y. Zhang, R. Lu, *J. Mater. Chem. C* **2015**, *3*, 1018; e) Q. Sun, W. Liu, S. Ying, L. Wang, S. Xue, W. Yang, *RSC Adv.* **2015**, *5*, 73046.
- [6] H. Yu, P. Tian, N. Han, M. Li, M. Wang, *Chem. Asian J.* **2023**, *18*, e202300712.
- [7] Y. Li, S. Liu, H. Ni, H. Zhang, H. Zhang, C. Chuah, C. Ma, K. S. Wong, J. W. Y. Lam, R. T. K. Kwok, J. Qian, X. Lu, B. Z. Tang, *Angew. Chem. Int. Ed.* **2020**, *59*, 12822–12826.
- [8] Deposition numbers 2360491 (for **An-1**), 2360492 (for **An-2**), and 2360493 (for **An-3**) contain the supplementary crystallographic data for this paper. These data are provided free of charge by the joint Cambridge Crystallographic Data Centre and Fachinformationszentrum Karlsruhe Access Structures service.

Manuscript received: August 1, 2024

Revised manuscript received: September 2, 2024

Accepted manuscript online: September 3, 2024

Version of record online: October 26, 2024